

# Ziyang Song

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## CONTACT INFORMATION

Stocker Center 313  
School of Electrical Engineering and Computer Science  
Ohio University  
Athens, Ohio, United States

ziyangs@ohio.edu  
Website  
LinkedIn  
Google Scholar

## RESEARCH INTERESTS

My research focuses on AI for health, with the goal of developing trustworthy and translational AI to address biomedical challenges.

1. **Trustworthy LLMs in Healthcare.** I develop methods to improve explainability, uncertainty quantification, and risk control for biomedical LLMs and multi-agent clinical systems.
2. **Foundation Models for Longitudinal Health Data.** I design foundation models for irregular and longitudinal healthcare data, including electronic medical records, laboratory measurements, and biosignals.
3. **Interpretable AI for Population Health.** I develop probabilistic models and scalable statistical inference methods to learn interpretable representations from electronic health records and genotype data, supporting phenotyping, incidence prediction, and gene discovery.
4. **Generative AI for Biology.** I develop foundation models for irregular single-cell RNA-seq time-series data and study domain adaptation methods that leverage unlabeled data.

## FULL-TIME PROFESSIONAL EXPERIENCE

**Ohio University**  
*School of Electrical Engineering and Computer Science*  
*Assistant Professor*

Athens, OH, United States  
Aug. 2025 – Present

## EDUCATION

**McGill University**  
*Ph.D. in Computer Science*

- **Affiliation:** Mila – Quebec AI Institute
- **Advisor:** Yue Li

Montreal, QC, Canada  
Sep. 2020 – May 2025

**Concordia University**  
*M.Sc. in Quality Systems Engineering*

- **Advisor:** Nizar Bouguila

Montreal, QC, Canada  
Sep. 2018 – Nov. 2019

**University of Shanghai for Science and Technology**  
*B.Eng. in Computer Science*

Shanghai, China  
Sep. 2014 – Jun. 2018

## PUBLICATIONS

Note: <sup>†</sup> equal contribution; \* corresponding authors.

### Conference Papers:

1. **Ziyang Song**, Qincheng Lu, Hao Xu, He Zhu, David Buckeridge, Yue Li. TimelyGPT: Extrapolatable Transformer Pre-training for Long-term Time-Series Forecasting in Healthcare. *ACM-BCB*. 2024. **Rising Star Award**.
2. Zilong Wang, Ruohan Wang, **Ziyang Song**, David Buckeridge, Yue Li. MixEHR-Nest: Identifying Subphenotypes within Electronic Health Records through Hierarchical Guided-Topic Modeling. *ACM-BCB*. 2024.
3. **Ziyang Song**, Qincheng Lu, He Zhu, David L. Buckeridge, Yue Li. Bidirectional Generative Pre-training for Improving Healthcare Time-series Representation Learning. *Machine Learning for Healthcare Conference (MLHC), PMLR*. 2024.
4. **Ziyang Song**, Yuanyi Hu, Aman Verma, David L. Buckeridge, Yue Li. Automatic Phenotyping by a Seed-guided Topic Model. *KDD*. 2022. **Health Day Best Paper Award**.

5. **Ziyang Song**, Xavier Sumba Toral, Yixin Xu, Aihua Liu, Liming Guo, Guido Powell, Aman Verma, David Buckeridge, Ariane Marelli, Yue Li. Supervised Multi-Specialist Topic Model With Applications on Large-Scale Electronic Health Record Data *ACM-BCB*. 2021.
6. **Ziyang Song**, Ornela Bregu, Samr Ali, Nizar Bouguila. Variational inference of finite asymmetric gaussian mixture models. *IEEE SSCI*. 2019.
7. **Ziyang Song**, Samr Ali, Nizar Bouguila. Bayesian learning of infinite asymmetric gaussian mixture models for background subtraction. *ICIAR*. 2019.

#### Journal Papers:

1. Ziqi Yang<sup>†</sup>, **Ziyang Song**<sup>†</sup>, Shadi Zabad, Marc-André Legault, Yue Li. PheCode-guided multi-modal topic modeling of electronic health records improves disease incidence prediction and GWAS discovery from UK Biobank. *Briefings in Bioinformatics*. 2026.
2. **Ziyang Song**, Qincheng Lu, He Zhu, David Buckeridge, Yue Li. TrajGPT: Irregular Time-Series Representation Learning of Health Trajectory. *IEEE Journal of Biomedical and Health Informatics (J-BHI)*. 2025.
3. **Ziyang Song**, Qincheng Lu, Hao Xu, Ziqi Yang, He Zhu, David Buckeridge, Yue Li. TimeLyGPT: Extrapolatable Transformer Pre-training for Long-term Time-Series Forecasting in Healthcare. *Health Information Science and Systems*. 2025.
4. Yuesong Zou, Ahmad Pesaranghader, **Ziyang Song**, Aman Verma, David Buckeridge, Yue Li. Modeling electronic health record data using an end-to-end knowledge-graph-informed topic model. *Scientific Reports*. 2022.
5. **Ziyang Song**, Samr Ali, Nizar Bouguila. Bayesian inference for infinite asymmetric gaussian mixture with feature selection. *Soft Computing*. 2021.
6. **Ziyang Song**, Samr Ali, Nizar Bouguila. Background subtraction using infinite asymmetric Gaussian mixture models with simultaneous feature selection. *IET Image Processing*. 2020.
7. **Ziyang Song**, Samr Ali, Nizar Bouguila, Wentao Fan. Nonparametric hierarchical mixture models based on asymmetric gaussian distribution. *Digital Signal Processing*. 2020.

#### Preprints & Under Submission:

1. Yu Gu, Zijun Yu, Chi-Kuang Yeh, Xinyu Wang, **Ziyang Song**<sup>\*</sup>. Severity-Controlled Prediction Sets for Medication Recommendation. Submitted to *NeurIPS 2026*, under review.
2. Yiran Ding, Yixuan Li, **Ziyang Song**<sup>\*</sup>. scUFC: Single-Cell Unsupervised Fine-Tuning via Clustering for Foundation Models. Submitted to *NeurIPS 2026*, under review.
3. **Ziyang Song**<sup>\*</sup>, Lia Shen, Yixuan Li, Qincheng Lu, Weihao Li, Haoran Xu, He Zhu, Xinyu Wang, Yue Li. SMI: Semantic Medical ID for Hierarchy-Aware Concept Representation. Submitted to *NeurIPS 2026*, under review.
4. Xinyu Wang, Yixuan Li, Hanwei Wu, Qincheng Lu, Chi-Kuang Yeh, Xiao-Wen Chang, **Ziyang Song**<sup>\*</sup>. Patients-like-me: A Variational LM-GNN Framework for Explainable Clinical Prediction. Submitted to *NeurIPS 2026*, under review.
5. Ruilin Wang, Bo-Hong Wang, Elizabeth Kourbatski, Hegang Chen, **Ziyang Song**, Gilles Boire, Marie Hudson, Yue Li. SciKit-Agent: Iterative Agentic Refinement for Tabular ML Pipelines. Submitted to *ICML 2026*, under review.
6. Jun Bai, **Ziyang Song**, Yue Li. FedEHR-GEN: Federated Synthetic Time-Series EHR Generation via Latent Space Alignment and Distribution-Aware Aggregation. Submitted to *KDD 2026*, under review.

INVITED TALKS &  
PRESENTATIONS

**Invited Talks**

|                                          |                                                                                   |           |
|------------------------------------------|-----------------------------------------------------------------------------------|-----------|
| OSC Research Symposium & NSF AI workshop | <i>Toward Trustworthy LLMs in Healthcare with Medical Ontologies.</i>             | Apr. 2026 |
| Global AI Youth Forum                    | <i>Toward Trustworthy LLMs in Healthcare with Medical Ontologies.</i>             | Mar. 2026 |
| Northern Arizona University              | <i>Representation Learning of Electronic Health Records.</i>                      | Mar. 2025 |
| San José State University                | <i>Representation Learning of Electronic Health Records.</i>                      | Mar. 2025 |
| Ohio University                          | <i>Representation Learning of Electronic Health Records.</i>                      | Feb. 2025 |
| South Dakota State University            | <i>Representation Learning of Electronic Health Records.</i>                      | Feb. 2025 |
| University of Texas at El Paso           | <i>Representation Learning of Electronic Health Records.</i>                      | Feb. 2025 |
| Microsoft Research Asia                  | <i>Time-series Transformer for Representation Learning and Health Forecasting</i> | Nov. 2024 |

**Guest Lectures**

|                   |                                                              |           |
|-------------------|--------------------------------------------------------------|-----------|
| Ohio University   | <i>Generative AI for Healthcare.</i>                         | Nov. 2025 |
| McGill University | <i>Time-series Transformer in Electronic Health Records.</i> | Nov. 2024 |
| McGill University | <i>Time-series Transformer in Electronic Health Records.</i> | Nov. 2023 |

**Oral Presentations**

|                                                                      |                                                                                                                |           |
|----------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|-----------|
| Canadian Society for Epidemiology and Biostatistics (CSEB)           | <i>AI-driven approach for computational disease phenotyping within the CARTaGENE cohort.</i>                   | Aug. 2025 |
| ACM BCB                                                              | <i>TimelyGPT: Extrapolatable Transformer Pre-training for Long-term Time-Series Forecasting in Healthcare.</i> | Nov. 2024 |
| Scientific Meeting of the Canadian Translational Geroscience Network | <i>AI-driven approach for computational phenotyping with CARTaGENE cohort.</i>                                 | Sep. 2024 |
| Tokyo Symposium on Genomic Medicine, Therapeutics and Health         | <i>Tutorial: Phenotyping and PheWAS using MixEHR-Seed.</i>                                                     | Apr. 2024 |
| ISMB GLBIO                                                           | <i>MixEHR-SAGE: A seed-guided topic model to improve phenotyping for PheWAS analysis in UK Biobank data.</i>   | May. 2023 |
| KDD                                                                  | <i>Automatic phenotyping by a seed-guided topic model.</i>                                                     | Aug. 2022 |
| ACM BCB                                                              | <i>Supervised multi-specialist topic model with applications on large-scale electronic health record data.</i> | Aug. 2021 |

## Poster Presentations

|                                                    |                                                                                                            |           |
|----------------------------------------------------|------------------------------------------------------------------------------------------------------------|-----------|
| Global AI Youth Forum                              | <i>Toward Trustworthy LLMs in Healthcare with Medical Ontologies.</i>                                      | Mar. 2026 |
| NeurIPS 2025 Responsible Foundation Model Workshop | <i>SMI: Semantic Medical ID for Hierarchy-Aware Concept Representation.</i>                                | Nov. 2025 |
| NeurIPS 2024 TSALM Workshop                        | <i>TrajGPT: Healthcare Time-Series Representation Learning for Trajectory Prediction.</i>                  | Dec. 2024 |
| MLHC                                               | <i>Bidirectional generative pre-training for improving healthcare time-series representation learning.</i> | Aug. 2024 |
| 50th Anniversary of SOCS at McGill University      | <i>MixEHR-Seed: Automatic phenotyping by a seed-guided topic model.</i>                                    | Oct. 2022 |
| ICIAR                                              | <i>Bayesian learning of infinite asymmetric gaussian mixture models for background subtraction.</i>        | Aug. 2019 |

## AWARDS, GRANTS, AND FUNDING

### As Principal Investigator (Aug. 2025 onwards)

|                                     |             |           |
|-------------------------------------|-------------|-----------|
| NVIDIA Academic Grant Program Award | <i>Gift</i> | Dec. 2025 |
|-------------------------------------|-------------|-----------|

### Prior to Principal Investigator Role (Before Aug. 2025)

|                                                |                    |                     |
|------------------------------------------------|--------------------|---------------------|
| FRQNT Provincial Doctoral Research Scholarship | <i>Scholarship</i> | May. 2023–Sep. 2025 |
| ACM BCB Rising Star Award                      | <i>Award</i>       | Nov. 2024           |
| KDD Health Day Best Paper Award                | <i>Award</i>       | Aug. 2022           |

## TEACHING EXPERIENCE

### Ohio University

|                   |                                                  |             |
|-------------------|--------------------------------------------------|-------------|
| <b>Instructor</b> |                                                  | Fall 2026   |
|                   | <i>AI 5010/4010 Foundations of Deep Learning</i> |             |
| <b>Instructor</b> |                                                  | Spring 2026 |
|                   | <i>AI 3300 Statistical Learning</i>              |             |
| <b>Instructor</b> |                                                  | Fall 2025   |
|                   | <i>AI 5010/4010 Foundations of Deep Learning</i> |             |

### McGill University

|                           |                                                             |             |
|---------------------------|-------------------------------------------------------------|-------------|
| <b>Teaching Assistant</b> |                                                             | Winter 2023 |
|                           | <i>COMP 551 Applied Machine Learning</i>                    |             |
| <b>Teaching Assistant</b> |                                                             | Fall 2022   |
|                           | <i>COMP 565 Machine Learning in Genomics and Healthcare</i> |             |

## INTERNSHIP EXPERIENCE

|                                                                                                                                                                     |                      |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| <b>Centre Hospitalier Universitaire Sainte-Justine (CHU-SJ)</b>                                                                                                     | Montreal, QC, Canada |
| <i>Research Intern</i>                                                                                                                                              | Jun. 2023–Dec. 2024  |
| Developed a clinical decision-support system using a probabilistic model to identify high-risk patients who may be missed by existing rule-based screening methods. |                      |
| <b>Nanyang Technological University</b>                                                                                                                             | Singapore            |
| <i>International Research Intern</i>                                                                                                                                | Jan. 2020–Jun. 2020  |
| Developed probabilistic models with stochastic optimization to analyze retail data, improving personalized product recommendations and targeted promotions.         |                      |

SUPERVISION &  
MENTORING

**Ohio University (Assistant Professor)**

**Ph.D. Students**

|                    |                   |
|--------------------|-------------------|
| Xuanlu Zhang       | Sep. 2026–Present |
| Shunning Liang Tan | Sep. 2026–Present |

**Undergraduate Students**

|                                                                                     |                   |
|-------------------------------------------------------------------------------------|-------------------|
| Sheil Iyer                                                                          | Sep. 2025–Present |
| <i>Topic/Output:</i> Agent with memory to unify email, Calendar, Notion, and Slack. |                   |

**Research Assistant Students**

|                                                                                   |                     |
|-----------------------------------------------------------------------------------|---------------------|
| Hanwei Wu (M.Sc., McMaster University)                                            | May. 2025–Present   |
| <i>Topic/Output:</i> Graph explainer for biomedical LLMs; Submitted to ICML 2026. |                     |
| Zihao Li (M.Sc., Northwestern University)                                         | Jul. 2025–Present   |
| <i>Topic/Output:</i> Semantic IDs in biomedical LLMs; NeurIPS Workshop 2025.      |                     |
| Yiran Ding (M.Sc., Brown University)                                              | Jul. 2025–Present   |
| <i>Topic/Output:</i> Representation clustering for biomedical LLMs.               |                     |
| Xiaohang Xie (B.S., OSU)                                                          | Jul. 2025–Present   |
| <i>Topic/Output:</i> Confidence calibration for biomedical LLMs.                  |                     |
| Yang Hu (B.S., OSU)                                                               | Jul. 2025–Present   |
| <i>Topic/Output:</i> Hyperbolic space for biomedical LLMs.                        |                     |
| Jiahe Shen (M.Sc., UW–Madison)                                                    | Jul. 2025–Dec. 2025 |
| <i>Topic/Output:</i> Semantic IDs in biomedical LLMs; NeurIPS Workshop 2025.      |                     |
| <i>Placement:</i> Research intern, MD Anderson Cancer Center.                     |                     |

**McGill University (Ph.D. Mentor)**

|                                                                                       |                     |
|---------------------------------------------------------------------------------------|---------------------|
| Ruilin Wang (M.Sc., McGill University)                                                | Sep. 2024–May. 2025 |
| <i>Topic/Output:</i> LLM agent. Submitted to ICML 2026.                               |                     |
| Ziqi Yang (B.S./M.Sc., McGill University)                                             | Sep. 2022–May. 2025 |
| <i>Topic/Output:</i> Expert-guided topic model for GWAS; Briefings in Bioinformatics. |                     |
| <i>Placement:</i> M.Sc. student, McGill University.                                   |                     |
| Ruohan Wang (B.S./M.Sc., McGill University)                                           | Sep. 2022–Sep. 2024 |
| <i>Topic/Output:</i> Expert-guided topic model; ACM-BCB 2024.                         |                     |
| <i>Placement:</i> Ph.D. student, Brown University.                                    |                     |
| Hao Xu (B.S., McGill University)                                                      | May. 2023–Sep. 2023 |
| <i>Topic/Output:</i> Time-series Transformer; ACM-BCB 2024.                           |                     |
| <i>Placement:</i> Ph.D. student, University of Pennsylvania.                          |                     |
| Ziyu Zhao (B.S., McGill University)                                                   | May. 2023–Sep. 2023 |
| <i>Topic/Output:</i> Time-series Transformer.                                         |                     |
| <i>Placement:</i> M.Sc. student, McGill University.                                   |                     |
| Yuanyi Hu (B.S., McGill University)                                                   | May. 2021–May. 2022 |
| <i>Topic/Output:</i> Expert-guided topic model; KDD 2022.                             |                     |
| <i>Placement:</i> M.Sc. student, Columbia University.                                 |                     |
| Yixin Xu (B.S., McGill University)                                                    | Sep. 2020–Sep. 2021 |
| <i>Topic/Output:</i> Multi-specialist topic model; ACM-BCB 2021.                      |                     |
| <i>Placement:</i> M.Sc. student, Duke University.                                     |                     |

## SERVICE

**Conference Reviewer**

|                |            |
|----------------|------------|
| ACM CAIS       | 2026       |
| NeurIPS        | 2024, 2026 |
| KDD            | 2024–2026  |
| ICML           | 2025–2026  |
| ICLR           | 2025–2026  |
| IJCAI          | 2025–2026  |
| CROSS-LLM      | 2025       |
| ACM SIGSPATIAL | 2025       |
| ML4H           | 2024       |

**Journal Reviewer**

|                                                |           |
|------------------------------------------------|-----------|
| Information Processing & Management            | 2026      |
| IEEE TNNLS                                     | 2026      |
| BMJ Digital Health & AI (expert peer reviewer) | 2025–2026 |
| TMLR                                           | 2025–2026 |
| Applied Intelligence                           | 2025      |
| Big Data Mining and Analytics                  | 2024–2025 |

**Grant/Proposal Reviewer**

|                                         |      |
|-----------------------------------------|------|
| ORAU Ralph E. Powe Junior Faculty Award | 2026 |
|-----------------------------------------|------|

**Ohio University**

|                                                                             |           |
|-----------------------------------------------------------------------------|-----------|
| CS/AI Assessment & Accreditation Committee member                           | 2025–2026 |
| Russ College of Engineering 3 Minute Thesis (3MT) Competition faculty judge | Feb. 2026 |

**McGill University and Mila**

|                                                           |      |
|-----------------------------------------------------------|------|
| Evaluation committee for Mila Supervision Request Process | 2024 |
|-----------------------------------------------------------|------|

## REFEREES

**Dr. Yue Li**, Associate Professor, School of Computer Science, McGill University  
*Email:* send.Li.D38D6CE548@interfoliodossier.com

**Dr. David Buckeridge**, Professor, School of Population and Global Health, McGill University  
*Email:* send.Buckeridge.BC61C28AEC2@interfoliodossier.com

**Dr. Chi-Kuang Yeh**, Assistant Professor, Department of Mathematics and Statistics, Georgia State University  
*Email:* cyeh@gsu.edu

**Dr. Archer Yi Yang**, Associate Professor, Department of Mathematics and Statistics, McGill University  
*Email:* send.Yang.BCEBBEFEC2@interfoliodossier.com